INCH-POUND
MIL-DTL-24643/52G
27 November 2023
SUPERSEDING
MIL-DTL-24643/52F
1 October 2009

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +194 DEGREES FAHRENHEIT (+90 DEGREES CELSIUS), 300 VOLTS, TYPE LSTPNW

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction (Non-Watertight)

First	-	Copper conductor, coated or uncoated 22 AWG, class K stranding.
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation 0.009 inch minimum average thickness. Telephone identification code shall be applied by method 3.
Fourth	-	Two conductors cabled together to form a pair. Maximum lay 2 inches. For LSTPNW-1½ only, three conductors, one white, one black, and one red, twisted together to form a triad having a left-hand lay length not greater than 2 inches.
Fifth	-	The required number of pairs (see table I), cabled together with a lay in accordance with 3.4.4 of MIL-DTL-24643. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outwards. Fillers may be used as necessary to form a firm, well rounded assembly.
Sixth	-	An optional binder.
Seventh	-	Cross-linked polyolefin jacket. See table I for thickness.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

Distribution Statement A. Approved for public release: distribution is unlimited.

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TABLE I. Details.

Military part no. M24643/52	Type and size	Number of pairs	Cable jacket thickness min. avg. (inch)	Overall diameter (inch)	
				min.	max.
-01UN	LSTPNW-1½	1½ 1/	0.040	0.217	0.235
-02UN	LSTPNW-3	3	0.040	0.287	0.310
-03UN	LSTPNW-5	5	0.040	0.338	0.365
-04UN	LSTPNW-10	10	0.040	0.435	0.470
-05UN	LSTPNW-15	15	0.040	0.490	0.530
-06UN	LSTPNW-20	20	0.040	0.532	0.575
-07UN	LSTPNW-30	30	0.050	0.629	0.680
-08UN	LSTPNW-40	40	0.050	0.708	0.765

NOTE:

1/ Size 1½ is composed of three conductors cabled to form a triad.

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	17.94
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	2,000
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to conductor	1,000
Conductor continuity	No failure

Group A:

Visual and dimensional	No failure

Group B:

Cold bending (cable)	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 392 °F [200 °C])	50
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700

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Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35

Group C:

Physicals (aged) (air oven)	
Insulation (extruded insulation only)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Shrinkage	No failure

Group D:

Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50

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Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type	Size	Feet (nominal)
LSTPNW	1½ through 10	2,500
LSTPNW	15 and 20	1,500
LSTPNW	30 and 40	1,000

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-068)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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